1 **(a)** $\frac{5}{8}$ 64% 0.65 $\frac{2}{3}$ **[B2]**

M1 for correct conversions to decimals seen: $0.666\ldots$, 0.64, 0.625

(b) $\frac{17}{20}$ **[B1]**

[Total Marks: 3]

2 **(a)** $\frac{7}{9}$ **[B1]**

(b) $\frac{36}{99}$ **[M1]**

$\frac{4}{11}$ **[A1]**

[Total Marks: 3]

3 $\frac{10}{3} \div \frac{14}{5}$ **[M1]**

$\frac{10}{3} \times \frac{5}{14}$ or $\frac{25}{21}$ **[M1]**

$\frac{5}{6}$ **[A1]**

[Total Marks: 3]

4 **(a)** $\frac{9}{20}$ **[B1]**

(b) 0.36 **[B1]**

(c) $\frac{204}{100}$ or $2\frac{4}{100}$ **[M1]**

$2\frac{1}{25}$ **[A1]**

[Total Marks: 4]

5 **(a)** $\frac{9}{20}$ **[B1]**

(b) 0.375 **[B1]**

(c) 0.4 $\frac{5}{12}$ 42% $\frac{3}{7}$ **[B2]**

If 0 scored, allow [M1] for converting at least two of the fractions or percentages to decimals correctly, e.g., $0.416\ldots$, 0.42, $0.428\ldots$

[Total Marks: 4]

6 **(a)** $\frac{75}{1000}$ or equivalent unsimplified fraction **[M1]**

$\frac{3}{40}$ **[A1]**

(b) 3.4 **[B1]**

(c) M1 for converting at least two of the other forms to decimals or percentages correctly (e.g. $\frac{5}{8} = 0.625$, $\frac{7}{11} = 0.636\ldots$, $62\% = 0.62$) **[M1]**

0.6 62% $\frac{5}{8}$ $\frac{7}{11}$ **[A1]**

[Total Marks: 5]

7 Length multiplier $= \frac{4}{3}$ **[B1]**

Area multiplier $= \frac{7}{6}$ **[B1]**

$\frac{4}{3} \times$ Width multiplier $= \frac{7}{6}$ oe **[M1]**

Width multiplier $= \frac{7}{6} \times \frac{3}{4} = \frac{7}{8}$ **[M1]**

$x = 12.5$ **[A1]**

[Total Marks: 5]

8 (a) 60 **[B3]**

M1 for $60\% = \frac{3}{5}$ or $\frac{1}{5} = 20\%$ seen

M1 for $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$ or $60\% - 20\% = 40\%$

M1 for $\frac{24}{2} \times 5$ or $\frac{24}{40} \times 100$

(b) 400 **[B3]**

M1 for $100\% - 30\% = 70\%$ or $0.7x$ seen

M1 for $\frac{3}{4} \times 0.7x = 210$ or $0.525x = 210$ or equivalent equation

[Total Marks: 6]

9 (a) Total number of counters $= x + (2x + 5) + (3x - 1) = 6x + 4$ **[M1]**

$\frac{2x+5}{6x+4} = \frac{35}{100}$ or $\frac{7}{20}$ **[M1]**

$20(2x + 5) = 7(6x + 4)$ leading to $40x + 100 = 42x + 28$ **[M1]**

36 **[A1]**

(b) Substituting $x = 36$ to find green counters $= 3(36) - 1 = 107$ and total counters $= 6(36) + 4 = 220$ **[M1]**

$\frac{107}{220} \times 100$ **[M1]**

$48\frac{7}{11}$ **[A1]**

[Total Marks: 7]

10 (a) $1000x = 245.4545\ldots$ and $10x = 2.4545\ldots$ or $100x = 24.5454\ldots$ and $x = 0.24545\ldots$ **[M1]**

$990x = 243$ or $99x = 24.3$ **[M1]**

$\frac{27}{110}$ **[A1]**

(b) Identifies that the denominator $42 = 2 \times 3 \times 7$ (or states that 42 has prime factors other than 2 and 5) **[B1]**

Terminating decimals can only have prime factors of 2 and/or 5 in their denominator. Since the fraction is in its simplest form, the factors of 3 and 7 do not cancel out, so it must recur. **[B1]**

(c) $\frac{10x+y}{99} = \frac{k}{11}$ or $\frac{9k}{99}$ (or states that the two-digit number xy is a multiple of 9) **[M1]**

9 **[A1]**

[Total Marks: 7]

11 (a) States appropriate pair of equations to eliminate the recurring part, e.g., $100x = 27.77\ldots$ and $10x = 2.77\ldots$ **[M1]**

Subtracts to get $90x = 25$ leading to $x = \frac{25}{90}$, and correctly simplifies this to $\frac{5}{18}$ **[A1]**

(b) (i) 36 **[B3]**

If 0 scored, allow [M1] for calculating the fraction or percentage of the whole school for either boys or girls, e.g., $\frac{2}{5} \times 45\%$ or $\frac{3}{5} \times 30\%$

Allow [M1] for evaluating either as 18%

(ii) 400 **[B2]**

If 0 scored, allow [M1] for a correct algebraic setup or division, e.g., $\frac{144}{0.36}$ or $36\% = 144$

[Total Marks: 7]

12 (a) $x - \frac{1}{3}x + 12$ or $\frac{2}{3}x$ **[M1]**

$\frac{2}{3}x + 12$ **[A1]**

(b) Expression after 2nd stop: $\frac{3}{4}\left(\frac{2}{3}x + 12\right)$ or $0.75\left(\frac{2}{3}x + 12\right)$ **[M1]**

Expands correctly to $\frac{1}{2}x + 9$ **[M1]**

Equates to final condition: $\frac{1}{2}x + 9 = x - 15$ **[M1]**

48 **[A1]**

(c) Final number of passengers $= 48 - 15 = 33$ **[B1]**

$\frac{33}{48} \times 100$ or $\frac{11}{16} \times 100$ or $\frac{275}{4}$ **[M1]**

$68\frac{3}{4}$ **[A1]**

[Total Marks: 9]

13 (a) 315 **[B1]**

(b) 27.5 **[B2]**

M1 for $\frac{3}{8} \times 100 = 37.5\%$ or for $100 - (35 + 37.5)$

(c) $\frac{9}{160}$ **[B2]**

M1 for $\frac{15}{100} \times \frac{3}{8}$ or $\frac{47.25}{840}$

[Total Marks: 5]

14 (a) Multiplier for 1st hour: $1 - 0.24 = 0.76$ (or 76%) **[B1]**

Multiplier for 2nd hour: $1 - \frac{2}{5} = \frac{3}{5} = 0.6$ (or 60%) **[B1]**

$0.76 \times 0.6 \times V = 45.6$ or $0.456V = 45.6$ **[M1]**

100 **[A1]**

(b) Total lost $= 100 - 45.6 = 54.4$ or $1 - 0.456 = 0.544$ **[M1]**

$\frac{68}{125}$ **[A1]**

[Total Marks: 6]

15 (a) $100\% - 45\% - 37.5\% = 17.5\%$ oe **[M1]**

$\frac{140}{0.175}$ oe **[M1]**

800 **[A1]**

(b) Junior students $= \frac{3}{8} \times 800 = 300$ **[M1]**

$1.5G + G = 300$ oe **[M1]**

120 **[A1]**

[Total Marks: 6]

16 (a) Converts at least two of the non-decimals to decimals correctly (e.g. 0.71 and 0.75) **[M1]**

$71\%, 0.714, \frac{5}{7}, \frac{3}{4}$ **[A1]**

(b) $\frac{1}{2}\left(\frac{2}{5}+\frac{5}{8}\right)$ or equivalent **[M1]**

$\frac{41}{80}$ **[A1]**

(c) $9x - 4x = 35$ or states the multiplier is $35 \div 5 = 7$ **[M1]**

$\frac{28}{63}$ **[A1]**

[Total Marks: 6]

17 (a) $\frac{2}{5} = 40\%$ or 0.4, and total percentage of Sales and Production $= 40\% + 28\% = 68\%$ **[M1]**

$100\% - 68\% = 32\%$ and equating $0.32 \times$ Total $= 24$ or equivalent **[M1]**

75 **[A1]**

(b) Number of women in Sales $= 0.60 \times 30 = 18$ **[M1]**

Number of women in Production $= \frac{3}{7} \times 21 = 9$ (or men $= 12$, so women $= 9$) **[M1]**

Total women $= 18 + 9 + 12 = 39$ **[M1]**

52 **[A1]**

[Total Marks: 7]

18 (a) 369 **[B2]**

If 0 scored, allow [M1] for 450×0.82 or 450×0.18

(b) 740 **[B3]**

If 0 scored, allow [M2] for $\frac{629}{1-0.15}$ or $\frac{629}{0.85}$

If 0 scored, allow [M1] for identifying that $85\% = \$629$

(c) 4 **[B2]**

If 0 scored, allow [M1] for a correct calculation showing the compound effect, e.g., $1.2 \times 0.8 = 0.96$ or choosing a starting value of 100 and reaching 96

[Total Marks: 7]